SOFTWARE DEVELOPMENT PROJECT SPECIFICATION

HEAVENSTAY WELLNESS RESORT MANAGEMENT SYSTEM (HSWRMS)

Course: ISP392 - Software Development Project

Team Structure: 6 Students / 6 Modules

Domain: Hospitality, Wellness Tourism, Healthcare, E-Commerce

Revision: Current ISP392 Project Baseline - September 2026

This document is a revised project specification adapted from the supplied baseline specification and aligned with the current HSWRMS scope agreed by the team: 6 modules, VNPay as the implemented payment gateway, Google and Facebook SSO, and a separate limited Manager role alongside Admin.

1. Problem Statement

Traditional Property Management Systems (PMS) are designed mainly for standard hotels where guests reserve rooms, check in, use basic services, and check out. A wellness resort operates differently because the guest experience is organized around a complete retreat journey: accommodation, retreat packages, sensitive health and dietary profiles, scheduled Spa/Therapy/Yoga sessions, personalized meals, extra services, consolidated billing, and management analytics.

HEAVENSTAY WELLNESS RESORT MANAGEMENT SYSTEM (HSWRMS) is proposed as an integrated information system for a fictional wellness resort scenario. The system must support the guest journey from registration and retreat booking through check-in, therapy scheduling, personalized F&B, checkout, payment, review, and reporting.

The main design challenge is not only to provide convenient screens, but to coordinate multiple operational resources and protect sensitive data. In particular, HSWRMS must:

- prevent double-booking of both Therapist and Treatment Room;
- protect health, allergy, dietary, and residence-identification data;
- allow each role to access only the information required for its work;
- aggregate Package, Spa, and F&B charges into one Guest/Villa Folio;
- prevent checkout while unpaid extra charges remain;
- support operational and management reporting without giving Manager full administrative authority.

2. Third-Party Tool Interfaces

HSWRMS integrates with the following external services.

2.1 VNPay Payment Gateway

Purpose: Process the retreat deposit and final checkout payment.

Main interactions:

- HSWRMS sends payment amount and transaction/order information to VNPay.
- VNPay returns payment result / payment confirmation to HSWRMS.
- HSWRMS records the confirmed transaction against the related booking or invoice.

2.2 Calendar and Notification Services

Google Calendar

Purpose: Synchronize confirmed Spa/Therapy/Yoga sessions to the Guest's personal calendar.

Email Notification Service (SendGrid or compatible provider)

Purpose:

- send email OTP for account verification;
- send therapy-session reminder approximately 1 hour before a confirmed session.

2.3 Authentication Providers

Google Identity and Facebook Login

Purpose: Provide Social Single Sign-On (SSO) in addition to the normal email/password authentication flow.

3. External Entity Specification (Actors)

1. Guest (Customer): Registers or signs in, maintains dietary/health profile, browses and books retreat packages, pays deposit, views booking and itinerary, schedules included Spa/Therapy sessions, selects meals, pays at checkout, and submits review/rating.
2. Receptionist (Front Desk): Views arrivals, verifies Guest information, collects residence-identification data, assigns a physical Villa at check-in, manages operational Villa status, manually books extra Spa services, reviews Guest Folio, generates consolidated invoice, and performs checkout.
3. Spa Therapist / Yoga Instructor: Views assigned daily schedule, accesses only treatment-relevant health notes of assigned Guests, performs sessions, and marks sessions Completed or No-Show.
4. F&B Staff / Chef: Views daily meal-preparation requirements, relevant allergy/dietary warnings, prepares meals, updates meal status, and handles extra a-la-carte orders without seeing unnecessary medical history.
5. Manager: Primarily views approved Dashboard/Report functions such as revenue, occupancy, therapist utilization, and system overview. Manager may receive limited delegated rights but must not have unrestricted Admin authority.
6. Admin: Manages system-level configuration, permission matrix, feature flags, audit information, staff/system administration, and approved master data.

External Systems

- VNPay Payment Gateway
- Google Calendar
- Email Notification Service
- Google Identity
- Facebook Login

4. Use Cases / User Stories

The project is organized into 6 independent but interconnected modules. Each student owns one module end-to-end and coordinates shared interfaces with the other modules.

Module 1 - Authentication & Sensitive Health Profile

- UC01 - Register, Verify Email & Login: As a Guest, I want to register, verify my email via OTP, log in securely, or use Google/Facebook SSO.
- UC02 - Manage Dietary & Health Profile: As a Guest, I want to create and update my Dietary & Health Profile with explicit, unchecked consent for sensitive-data collection.
- UC03 - Manage Staff Accounts & Roles: As an authorized administrator, I want to create, edit, deactivate staff accounts and assign roles.
- UC04 - Manage Master Data: As an authorized administrator, I want to manage approved master data such as Villa Categories, Spa Services, and Retreat Packages using appropriate wellness terminology.
- UC05 - Request Health Data Deletion: As a Guest, I want to request deletion/anonymization of my health and allergy data after my retreat has ended, subject to required audit/retention rules.

Module 2 - Retreat Package & Accommodation Booking

- UC06 - Browse Retreat Packages: As a Guest, I want to browse available Retreat Packages and filter them by wellness goal.
- UC07 - Book Retreat & Pay Deposit: As a Guest, I want to select a package, dates and Villa type, then pay a deposit securely through VNPay.
- UC08 - Check-In & Assign Villa: As a Receptionist, I want to view expected arrivals, verify the Guest, collect required residence-identification information, and assign a specific physical Villa.
- UC09 - Manage Villa Status: As a Receptionist, I want to manage physical Villa status such as Available, Occupied, Maintenance, and post-checkout cleaning state.
- UC10 - View Booking & Itinerary: As a Guest, I want to view my booking details and itinerary timeline.

Residence-compliance clarification for UC08: Required Guest identification data must be protected at rest, and authorized management must be able to export the daily residence-declaration report required by the project's residence-law research.

Module 3 - Spa & Therapy Scheduling Engine - CORE 1

- UC11 - Schedule Included Spa/Therapy Session: As a Guest with a valid retreat booking, I want to schedule an included Spa/Therapy/Yoga session by selecting an eligible date/time slot.
- UC12 - Match Therapist & Treatment Room: The system must confirm a slot only when both an eligible Therapist and an available Treatment Room exist for the same time interval.
- UC13 - View Therapist Schedule & Health Notes: As a Therapist, I want to view my assigned daily schedule and only the treatment-relevant health notes of assigned Guests.
- UC14 - Mark Session Completed / No-Show: As a Therapist, I want to update the outcome of an assigned session.
- UC15 - Manually Book Extra Spa Service: As a Receptionist, I want to manually book an extra Spa service for a Guest; the booking must use the same resource-conflict checks and the charge must be posted to the Guest/Villa Folio.

Supporting behavior: Confirmed sessions are synchronized to Google Calendar and receive an email reminder approximately 1 hour before the session.

Module 4 - Dietary F&B Management - CORE 2

- UC16 - Pre-select Daily Meals: As a Guest, I want to pre-select daily meals from options filtered by my allergy/dietary profile.
- UC17 - View Daily Meal Prep Dashboard: As Chef/F&B Staff, I want to view aggregated daily meal orders and relevant allergy warnings.
- UC18 - Update Meal Order Status: As Chef/F&B Staff, I want to change meal status from Preparing to Ready for Delivery.
- UC19 - Order Extra A-la-carte Item: As a Guest, I want to order an extra a-la-carte item and charge it to my Villa Folio.
- UC20 - Minimize Health Data for Chef: The system must expose only food-related allergy/dietary information to Chef/F&B Staff and must hide unrelated medical history.

Module 5 - Consolidated Checkout & Analytics

- UC21 - Generate Consolidated Invoice: As a Receptionist, I want to generate one invoice that consolidates remaining Package cost, Extra Spa services, and Extra F&B orders for the booking.
- UC22 - Process Final Payment & Update Villa: As a Receptionist, I want to process final payment through VNPay and update the Villa state after successful checkout.
- UC23 - Submit Review & Rating: As a Guest, I want to submit a review and rating after checkout.
- UC24 - View Revenue Dashboard: As a Manager, I want to view revenue breakdown by Package, Spa, and F&B.
- UC25 - Export Occupancy & Therapist Utilization: As a Manager, I want to export occupancy and therapist-utilization reports.

Checkout guard: UC22 may proceed only when all pending/unpaid Extra Spa and F&B charges have been settled.

Module 6 - System Administration & Permission Management

- UC26 - Configure Business Parameters: As an Admin, I want to configure approved system parameters such as refund percentage and cancellation window.
- UC27 - Toggle Feature Flags: As an Admin, I want to enable or disable approved functionality by module.
- UC28 - Configure Permission Matrix: As an Admin, I want to configure detailed permissions by screen/action/role.
- UC29 - Delegate Limited Admin Rights: As an Admin, I want to delegate a limited subset of administrative rights to Manager without granting full Admin authority.
- UC30 - View Sensitive-Data Audit Log: As an Admin, I want to review sensitive-data access events for governance and troubleshooting.
- UC31 - View System Overview Dashboard: As an Admin or authorized Manager, I want to view an overview of system operations and management indicators.

5. Business Rules (Strict Constraints)

1. BR-01 - Two-Dimensional Double-Booking Prevention: A Spa booking is valid only if a specific eligible Therapist and a specific Treatment Room are both available for the requested time slot. The system must reserve both resources consistently so concurrent requests cannot create conflicting bookings.
2. BR-02 - Consolidated Billing / Checkout Guard: A Guest cannot complete checkout while pending or unpaid Extra Spa or F&B charges remain.
3. BR-03 - Data Minimization and RBAC:
   - Therapists can view only health data relevant to physical treatment and only for assigned Guests.
   - Chef/F&B Staff can view only food-related allergy/dietary information required for meal preparation.
   - Receptionists must not see unnecessary sensitive Health/Diet data.
   - Access rules must be enforced in backend authorization and data-access logic, not only in the UI.

4. BR-04 - Folio Accounting: Extra Spa and F&B charges must be posted to the Guest/Villa Folio and linked to the related `Room_Booking_ID` so Module 5 can calculate the consolidated invoice.
5. BR-05 - Residence Data Protection: Residence-identification data collected during check-in must be protected at rest and accessed only by authorized roles.
6. BR-06 - Manager Least Privilege: Manager may view approved dashboards/reports and receive explicitly delegated limited rights, but must not inherit unrestricted Admin permissions.
7. BR-07 - Right to Deletion Timing: Guest health/allergy deletion or anonymization is available after the retreat has ended and must preserve any legally or operationally required audit evidence.

6. References & Domain Knowledge (Mandatory Research)

Students must research the following domain-specific requirements and reflect the findings in database design, authorization, UI behavior, and SRS business rules.

A. Legal Compliance & Data Protection

1. Vietnam Decree 356/2025/ND-CP on Personal Data Protection (as stated in the supplied project baseline)

Focus areas: sensitive health/biometric data and explicit consent.

Project application: Health/Diet data requires explicit consent, must not use default-checked consent, and must be protected appropriately.

2. Law on Residence 2020 (Luật Cư trú 2020)

Focus area: mandatory temporary-residence declaration for domestic and foreign Guests.

Project application: check-in collects required identification information; the data is protected at rest; authorized management can produce the daily residence-declaration report required by the project scope.

B. Hospitality & Wellness Domain Operations

3. Global Wellness Institute (GWI) Standards / Wellness Taxonomy

Focus area: taxonomy and terminology for wellness tourism.

Project application: Retreat Package master data should use wellness-domain terminology instead of generic hotel wording.

4. Guest Folio & Night Audit Practices (AHLEI-oriented research)

Focus area: accounting mechanisms in which point-of-sale services such as Spa and F&B post charges to a central Guest Folio.

Project application: Module 5 aggregates Package, Spa, and F&B charges using the booking as the primary business link.

C. Technical Synthesis

5. Encryption and Concurrency Research

Focus areas: encryption of sensitive data at rest, RBAC enforcement, transactional/concurrency controls, and prevention of Spa resource double-booking.

7. Module Integration Summary

The six modules form one end-to-end business flow:

```
Authentication & Health Profile -> Retreat Booking & Deposit -> Check-In & Villa Assignment -> Spa/Therapy + F&B Operations -> Guest/Villa Folio -> Consolidated Checkout & Analytics
```

Module 6 (System Administration & Permission Management) operates across the whole system to control permissions, configuration, feature flags, auditability, and management visibility.

The system-wide integrations are:

- M1 <-> M6: account roles and permission matrix;
- M2 / M5 <-> VNPay: deposit and final payment;
- M3 <-> Google Calendar / Email: session synchronization and reminder;
- M2 + M3 + M4 -> M5: package/service charges aggregated into the Folio and consolidated invoice.

End of Revised HSWRMS Project Specification